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Student Submissions — CO2_AT3_Coding Challenge

SIMATS Engineering • CSE • CSA0804 — Python Programming • 27-07-2026 • Category: CO2_AT3_Coding Challenge • Student Submissions Report

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Q1. String Character Frequency Counter

CODE

```
s = input("Enter a string: ").lower()

count = {}

for ch in s:
    if ch != " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1

for ch in sorted(count):
    print(ch, "→", count[ch])
```

INPUT

Hello World

OUTPUT

```
Enter a string: d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

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Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()

vowels = "aeiou"
vowel_count = 0
consonant_count = 0
```

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Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()

vowels = "aeiou"
vowel_count = 0
consonant_count = 0
vowel_freq = {}
cons_freq = {}

for ch in s:
    if ch.isalpha():
        if ch in vowels:
            vowel_count += 1
            vowel_freq[ch] = vowel_freq.get(ch, 0) + 1
        else:
            consonant_count += 1
            cons_freq[ch] = cons_freq.get(ch, 0) + 1

print("Vowels:", vowel_count)
print("Consonants:", consonant_count)

if vowel_freq:
    mv = max(vowel_freq, key=vowel_freq.get)
    print("Most frequent vowel:", mv, "(", vowel_freq[mv], ")")
else:
    print("Most frequent vowel: None")

if cons_freq:
    lc = min(cons_freq, key=cons_freq.get)
    print("Least frequent consonant:", lc, "(", cons_freq[lc], ")")
else:
    print("Least frequent consonant: None")

if consonant_count > 0:
    print("Vowel-to-Consonant Ratio:", round(vowel_count / consonant_count, 2))
else:
    print("Vowel-to-Consonant Ratio: Undefined")
```

INPUT Python Programming

OUTPUT
Vowels: 4
Consonants: 13
Most frequent vowel: o (2)
Least frequent consonant: y (1)
Vowel-to-Consonant Ratio: 0.31

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Q3. String Compression and Run-Length Encoding

CODE



439b19ff-bf54-4bf6-b636-ba9...



INPUT Python Programming

OUTPUT

```
Vowels: 4
Consonants: 13
Most frequent vowel: o ( 2 )
Least frequent consonant: y ( 1 )
Vowel-to-Consonant Ratio: 0.31
```

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Q3. String Compression and Run-Length Encoding

CODE

```
s = input()

if s == "":
    print("Compressed:")
    print("Decompressed:")
else:
    compressed = ""
    count = 1

    for i in range(1, len(s)):
        if s[i] == s[i - 1]:
            count += 1
        else:
            compressed += s[i - 1]
            if count > 1:
                compressed += str(count)
            count = 1

    compressed += s[-1]
    if count > 1:
        compressed += str(count)

    if len(compressed) < len(s):
        print("Compressed:", compressed)

        decompressed = ""
        i = 0
        while i < len(compressed):
            ch = compressed[i]
            i += 1
            num = ""
            while i < len(compressed) and compressed[i].isdigit():
                num += compressed[i]
                i += 1
            if num == "":
                decompressed += ch
            else:
                decompressed += ch * int(num)

        print("Decompressed:", decompressed)
    else:
        print("Compression not beneficial.")
    print("Original:", s)
```

INPUT aabbccdddee

OUTPUT

```
Compressed: a2b3c2d4e2
Decompressed: aabbccdddee
```



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OUTPUT

Compressed: a2b3c2d4e2
Decompressed: aabbccdddeee

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Q4. Advanced Text Statistics Analyser

CODE

```
import re

text = input()

stop_words = {"the", "is", "a", "an", "in", "of", "and", "to"}

paragraphs = 1 if text.strip() else 0
sentences = max(1, len(re.findall(r"[.!?]", text)))

words = re.findall(r"\b[a-zA-Z]+\b", text.lower())
word_count = len(words)

avg_word_length = round(sum(len(w) for w in words) / word_count, 2) if word_count else 0
avg_sentence_length = round(word_count / sentences, 2) if sentences else 0

freq = {}
for w in words:
    if w not in stop_words:
        freq[w] = freq.get(w, 0) + 1

top5 = sorted(freq.items(), key=lambda x: (-x[1], x[0]))[:5]

vowels = "aeiou"
syllables = 0
for w in words:
    c = 0
    prev = False
    for ch in w:
        if ch in vowels:
            if not prev:
                c += 1
            prev = True
        else:
            prev = False
    syllables += max(c, 1)

score = 206.835 - 1.015 * (word_count / sentences) - 84.6 * (syllables / word_count)

if score >= 90:
    level = "Very Easy"
elif score >= 80:
    level = "Easy"
elif score >= 70:
    level = "Fairly Easy"
elif score >= 60:
    level = "Standard"
elif score >= 50:
    level = "Fairly Difficult"
elif score >= 30:
    level = "Difficult"
else:
    level = "Very Confusing"

print("Words:", word_count, "| Sentences:", sentences, "| Paragraphs:", paragraphs)
print("Avg Word Length:", avg_word_length, "| Avg Sentence Length:", avg_sentence_length)
print("Top 5 Words:", ", ".join(f"{w}({c})" for w, c in top5))
print("Flesch Reading Ease:", round(score, 1), f"({level})")
```

OUTPUT

"C:\PyComp\PyC72\python.exe" is not recognized as an internal or external command,
operable program or batch file.

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